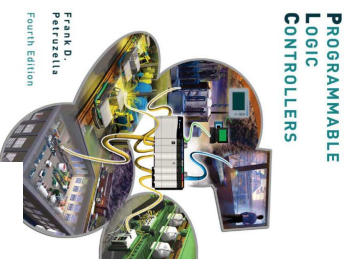


Programmable logic controllers and industrial controllers

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mailto:\\Stefan.Brock@put.poznan.pl



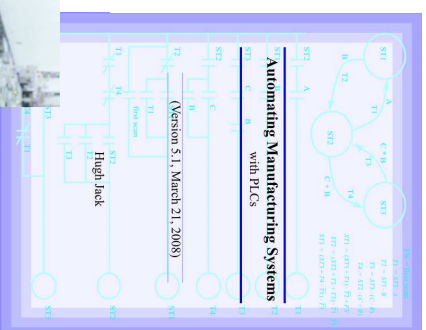
SIEMENS

SIMATIC

S7-1200 Programmable controller

System Manual

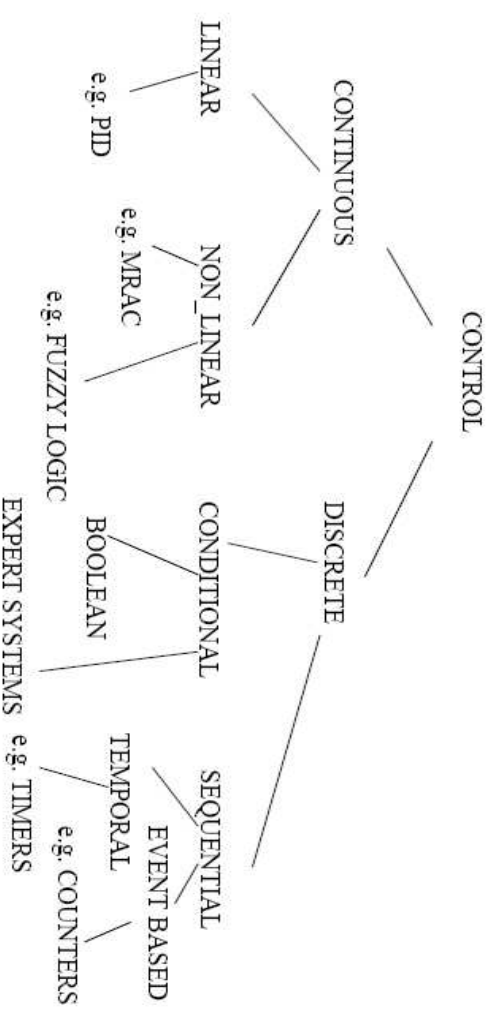
Position	Module
1	Power supply
2	STEP 7 programming
3	Industrial Ethernet
4	PS 307 5A
5	Power supply
6	Power supply
7	Power supply
8	Power supply
9	Power supply
10	Power supply
11	Power supply
12	Power supply
13	Power supply
14	Power supply
A	Power supply
B	Power supply
C	Power supply



Main topics

- Hardware of PLC
- Programming of PLC
- Sensors and actuators
- Digital control algorithms
- Selected topics on fieldbusses
- Introduction to SCADA and HMI

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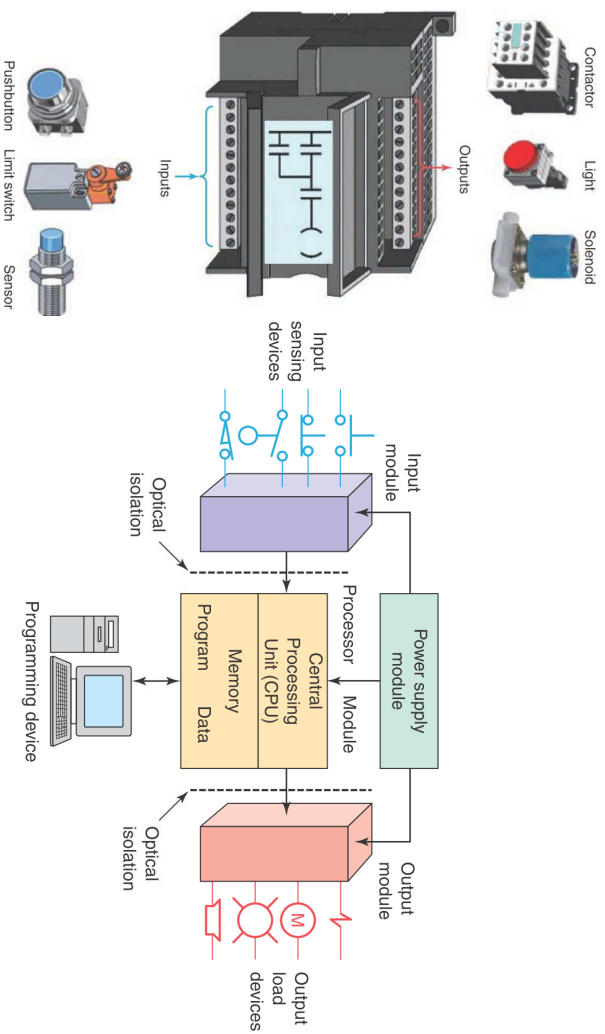
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PLC history

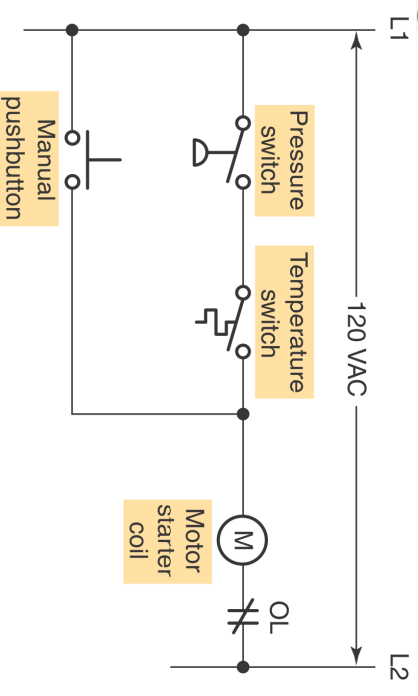
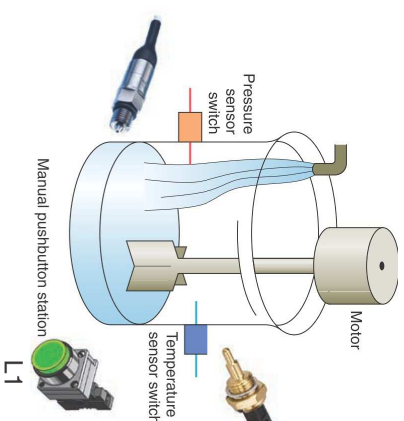
1968 General Motors established criteria for a programmable controller. The aim was to replace expensive and inflexible relay control systems. The main requirements are:

- working under industrial conditions
- simple programming by technicians and engineers
- reusable

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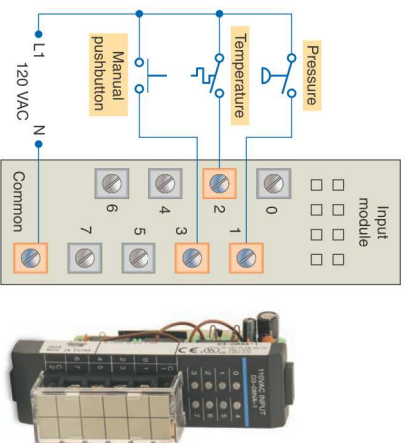
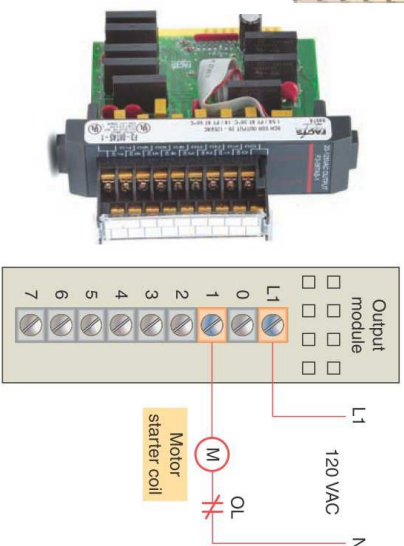


Figure 1-19 Typical wiring connections for a 120 VAC modular configured input module.
Source: Photo courtesy Automation Direct, www.automationdirect.com.



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Figure 1-20 Typical wiring connections for a 120 VAC modular configured output module.
Source: Photo courtesy Automation Direct, www.automationdirect.com.

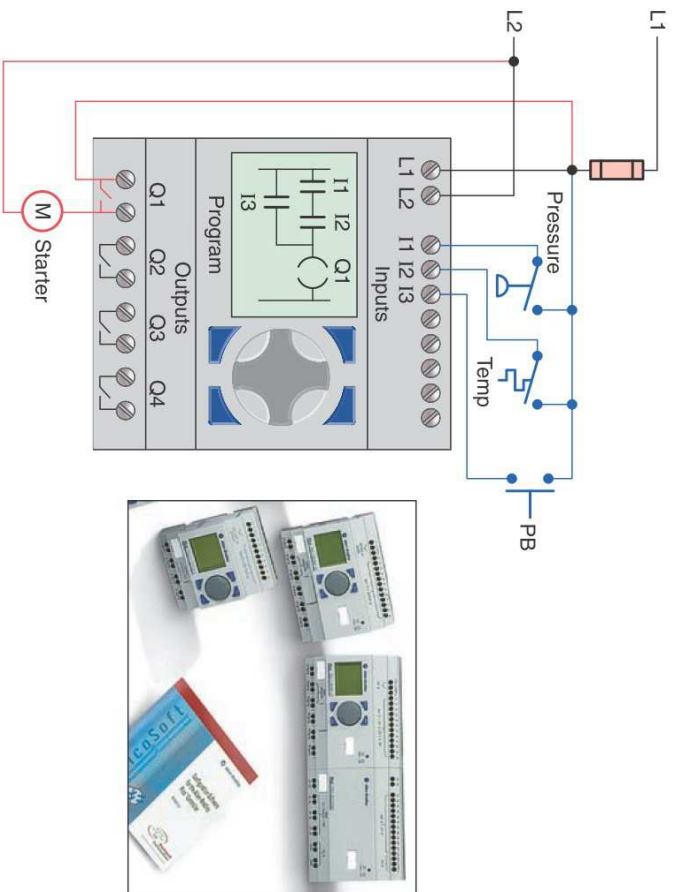


Figure 1-22 Typical wiring required to implement the process control scheme using a fixed PLC controller.

Source: Image Used with Permission of Rockwell Automation, Inc.

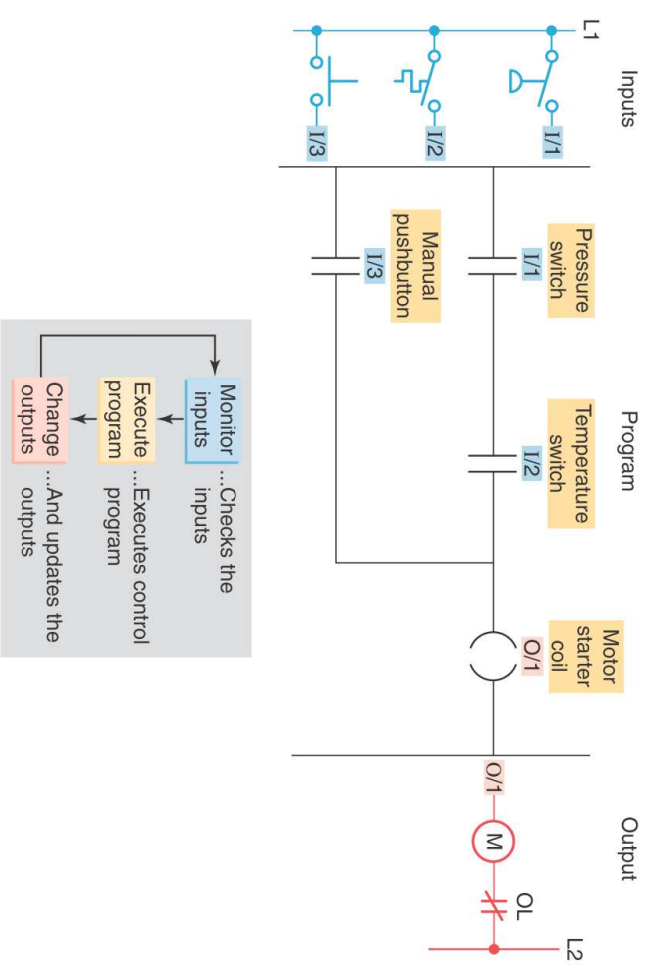


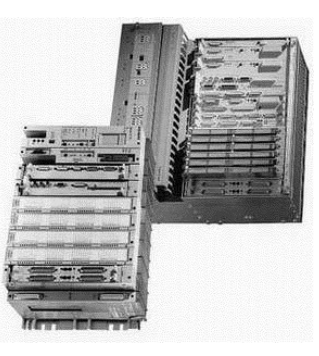
Figure 1-21 Process control PLC ladder logic program with typical addressing scheme.

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PLC, DCS, PAC i IPC

Programmable logic controllers

- Replace the relay control - introduced after the end of the 1960s.
- Sequential control - production line, sequence of operations
- Initially low memory and low computing power
- Now work in distributed industrial network systems

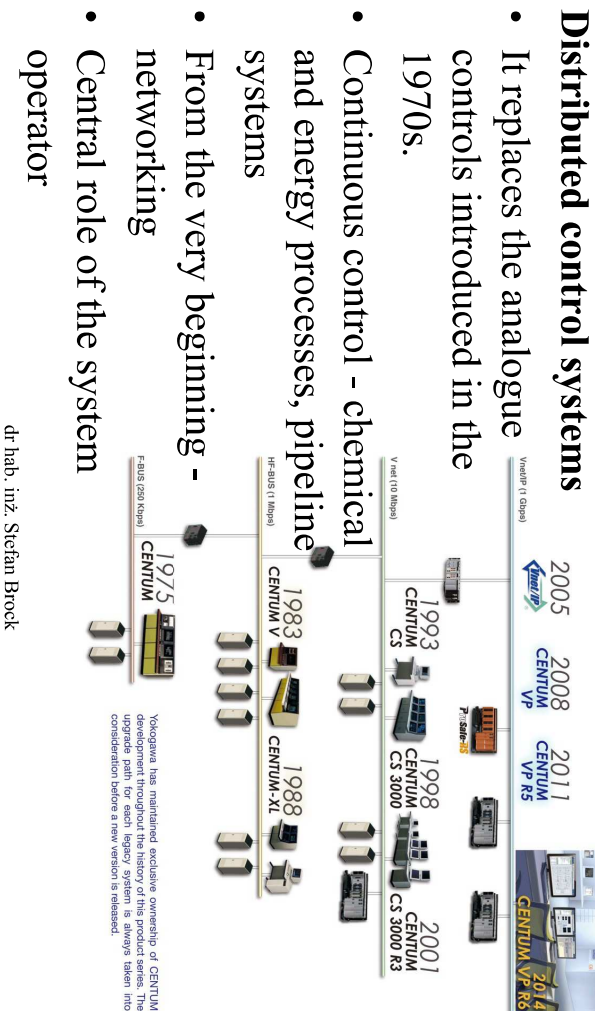


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PLC, DCS, PAC i IPC

CENTUM history Nearly 40 years and still counting



PLC, DCS, PAC i IPC

Industrial PCs

- From the 1990s onwards. emulation of the PLC in the PC environment
- Problems - environment, unstable operating system
- PC platform - powerful processors and memory, easy integration with HMI
- Real-time kernel



Industrial fanless PC



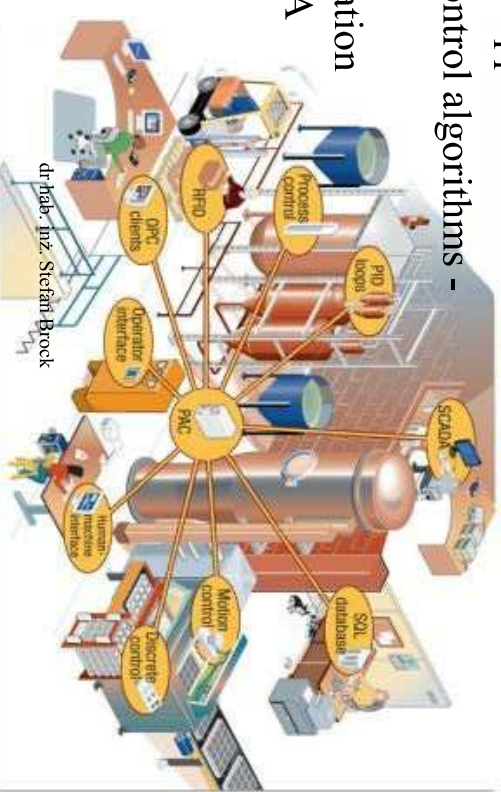
Industrial Panel PC

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PLC, DCS, PAC i IPC

Process Automation Controllers

- PLC successor, increased computing power and memory, wider field of application
- Advanced control algorithms - MPC, fuzzy
- Easy cooperation with SCADA



IEC 61131-3

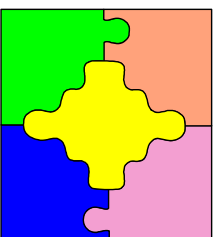
“The best thing that happened to industrial control”

Sugar Lantic on Automation Maillist

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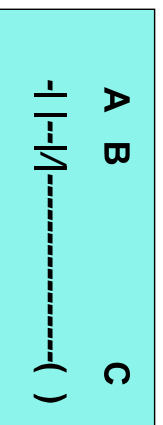
The 8 parts of the IEC 61131 Standard

- 1 General overview, definitions
- 2 Hardware
- 3 Programming Languages
- 4 User Guidelines
- 5 Messaging Service Specification
- 6 *Functional safety*
- 7 Fuzzy Logic
- 8 Application and Implementation of Programming Languages
- 9 Digital communication interface for small sensors and actuators (IO-Link)
- 10 PLC open XML exchange format for the export and import of IEC 61131-3 projects



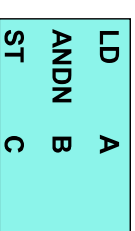
Ladder Diagram (LD)

- Standardized, rationalized set of relay ladder programming symbols
- Based on well-known North american style of programming, resembling US-type of electrical drawing standard



The IEC 61131-3 Programming Languages

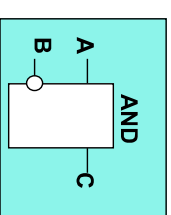
Instruction List



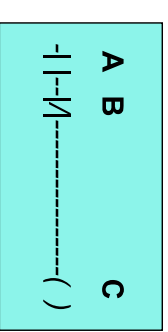
Structured Text



Function Block Diagram



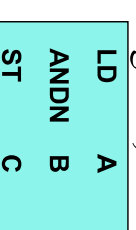
Ladder Diagram



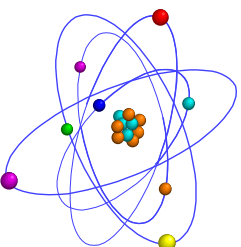
Instruction List (IL)



- Single Accumulator based execution model
- Based upon the German ‘Anweisungsliste’, A WL
- One operation such as storing a value in the accumulator register, is allowed per line



Structured Text (ST)



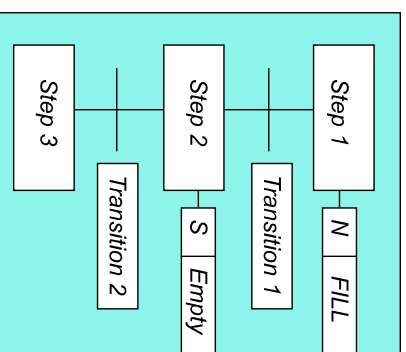
- High level language, block structured
- Syntax resembles PASCAL
- Complex statements and nested instructions possible
- Support for
 - Iteration loops (REPEAT-UNTIL; WHILE-DO)
 - Conditional execution (IF-THEN-ELSE; CASE)
 - Functions (SQRTO, SIN())

```
C := A AND NOT B
```

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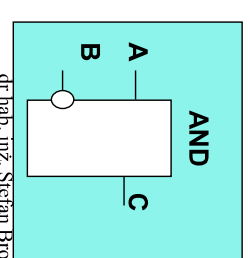
Sequential Function Chart, SFC

- Powerful graphical technique for describing the sequential behaviour of a control program
- Used to partition a control problem
- Shows overview, also suitable for rapid diagnostics
- The basic elements are **STEPS** with **ACTION BLOCKS** and **TRANSITIONS**
- Support for alternative and parallel sequences

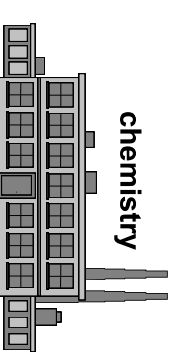


Function Block Diagram (FBD)

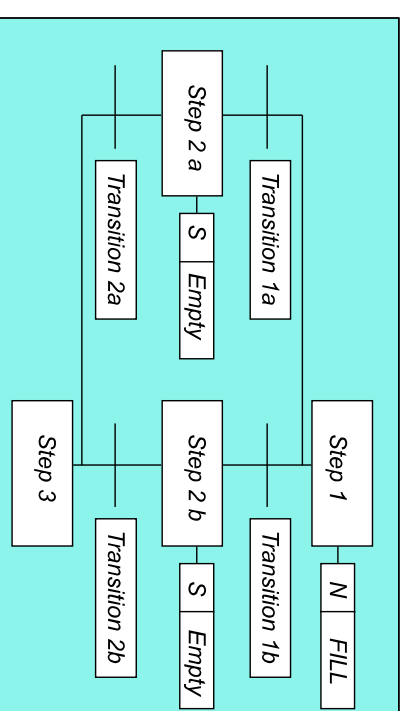
- Graphical language, widely used in Europe
- Allows program elements which appear as blocks to be "wired" together in a form analogous to a circuit diagram
- Used in many applications that involve the flow of information or data between control components



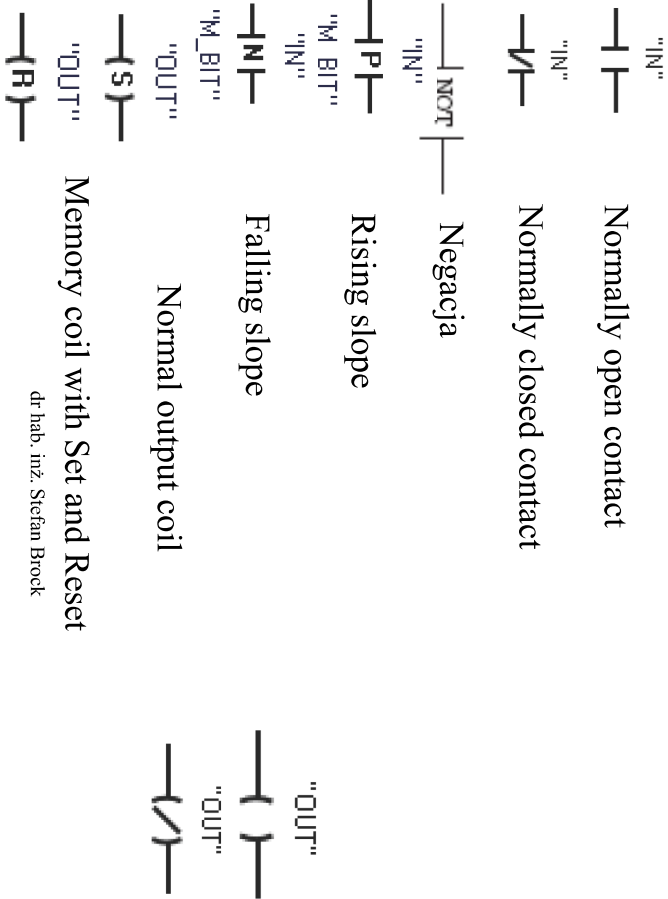
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SFC : alternative sequences

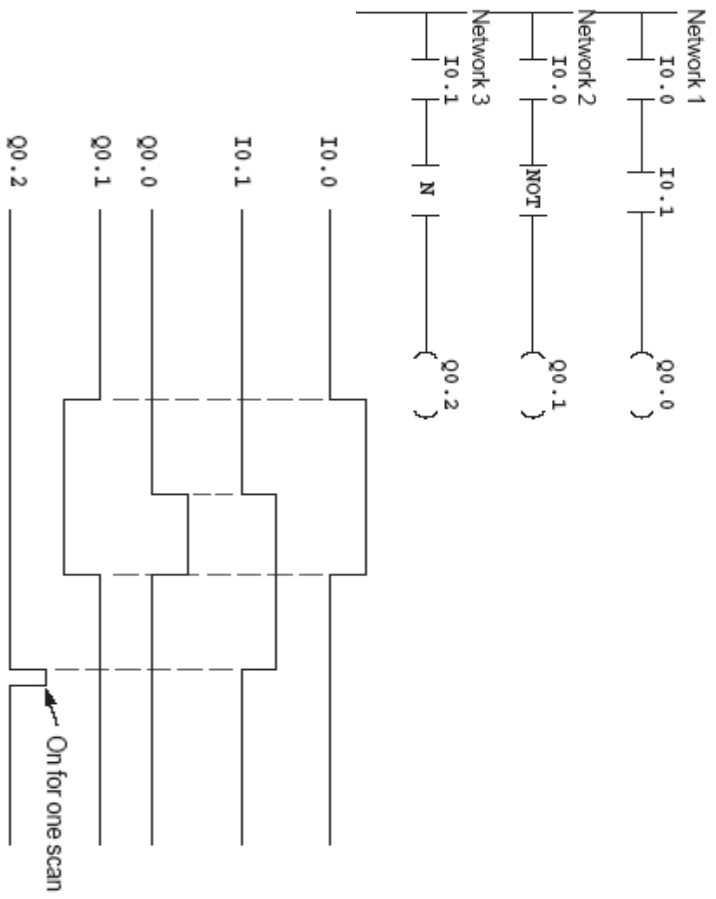
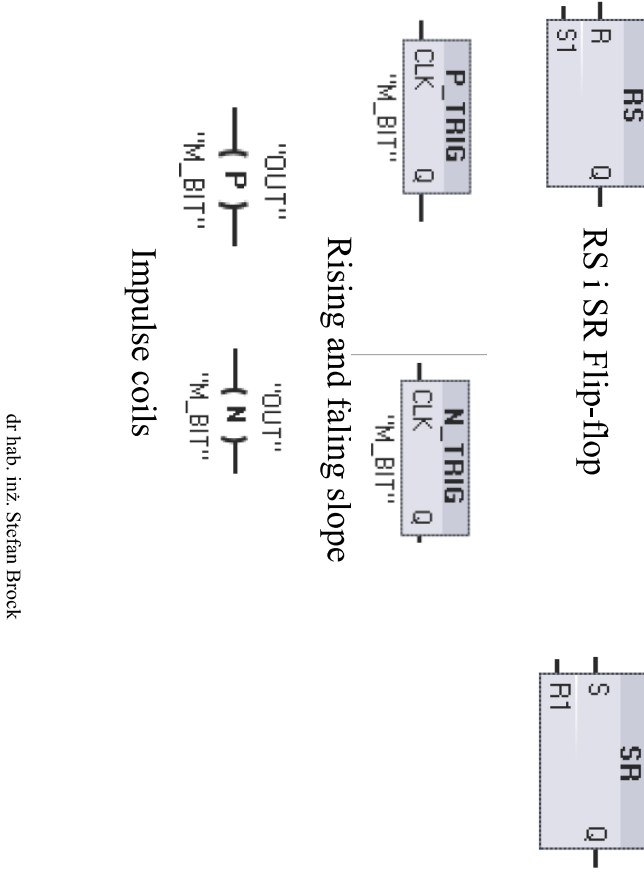


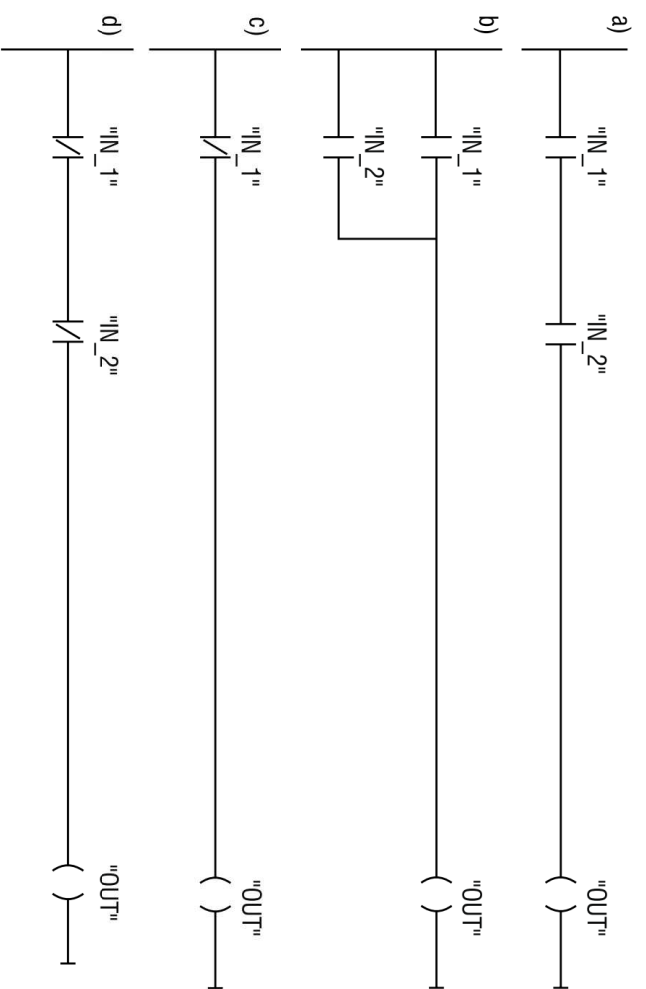
Elements of the LD language



Basic instructions		
General		
Bit logic operations		
— I —	— I —	Normally open contact – styk normalnie otwarty (sprawdzenie, czy argument jest w stanie wysokim)
— I —	— I —	Normally closed contact – styk normalnie zamknięty (sprawdzenie, czy argument jest w stanie niskim)
— I —	— NOT —	Invert result of logic operation – negacja funkcji logicznej
— I —	— < —	Output Coil – cewka wyjściowa, zapis wyniku funkcji logicznej do argumentu
— I —	— < —	Negated Coil – zanegowana cewka wyjściowa, zapis negacji wyniku funkcji logicznej do argumentu
— I —	— < —	Reset – warunkowe kasowanie bitu
— I —	— < —	Set – warunkowe ustawianie bitu
— I —	— < —	SET_BF (Set bit field) – warunkowe ustawienie zdefiniowanej grupy bitów, począwszy od argumentu
— I —	— < —	RESET_BF (Reset bit field) – warunkowe wyzerowanie zdefiniowanej grupy bitów, począwszy od argumentu
— I —	— < —	SR (Set/reset flip-flop) – przerzutnik z priorytetem na wejście kasujące
— I —	— < —	RS (Reset/set flip-flop) – przerzutnik z priorytetem na wejście ustawiające
— I —	— < —	Scan positive edge at operand – wykrywanie zbocza narastającego bitu wejściowego
— I —	— < —	Scan negative edge at operand – wykrywanie zbocza opadającego bitu wejściowego
— I —	— < —	Set operand on positive signal edge – ustawia argument w przypadku wystąpienia zbocza narastającego bitu RLO
— I —	— < —	Set operand on negative signal edge – ustawia argument w przypadku wystąpienia zbocza opadającego bitu RLO
— I —	— < —	P_TRIG (Set output on positive signal edge) – ustawia wyjście w przypadku wystąpienia zbocza narastającego bitu wejściowego
— I —	— < —	N_TRIG (Set output on negative signal edge) – ustawia wyjście w przypadku wystąpienia zbocza opadającego bitu wejściowego

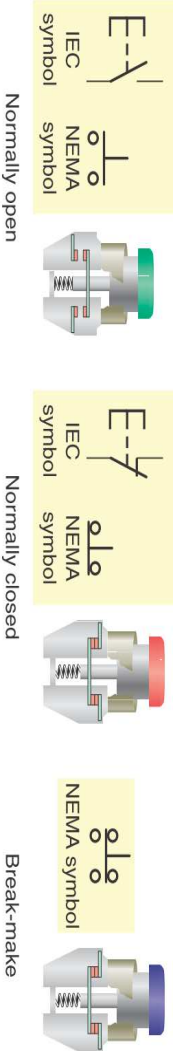
Elements of the LD language



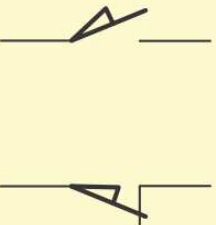


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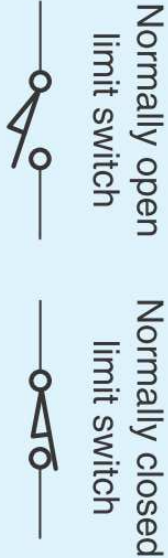
Buttons



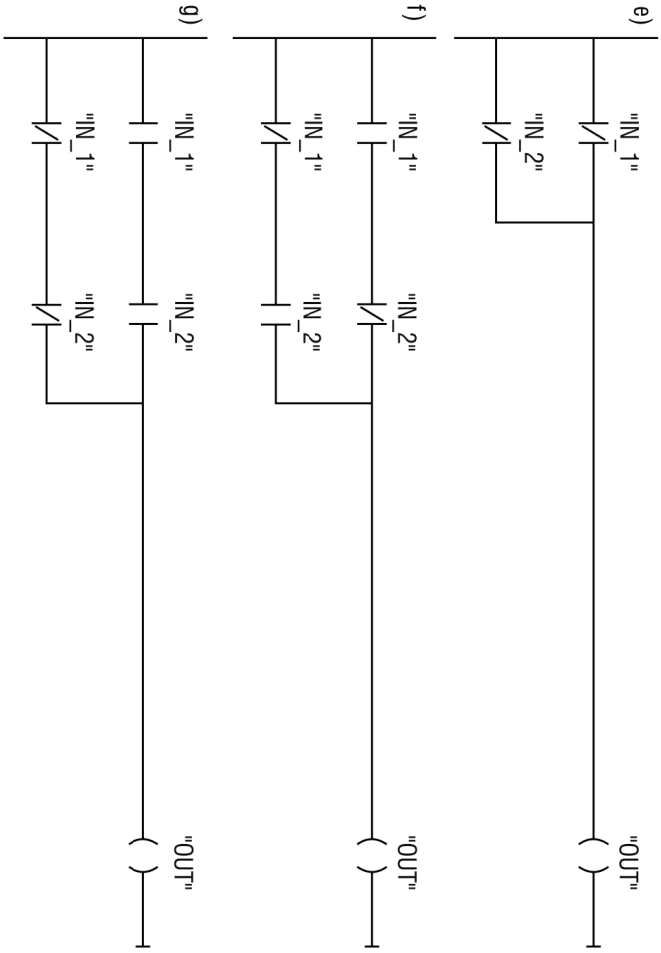
IEC symbols



NEMA symbols

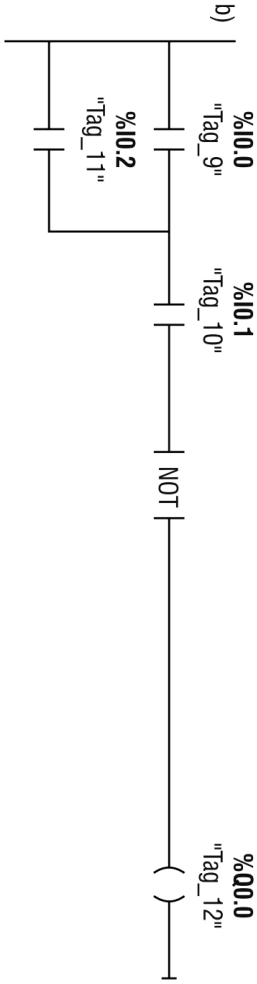


Contacts



Wykorzystano ilustracje z książki:
J. Kwaśniewski „Sterowniki S7-1200
w praktyce inżynierskiej”

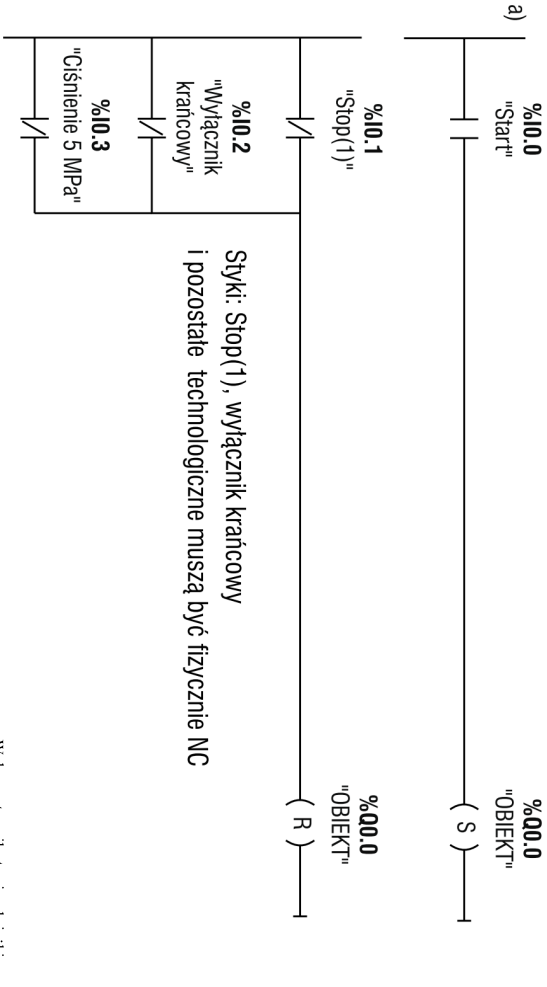
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Wykorzystano ilustracje z książki:
J. Kwaśniewski „Sterowniki S7-1200
w praktyce inżynierskiej”

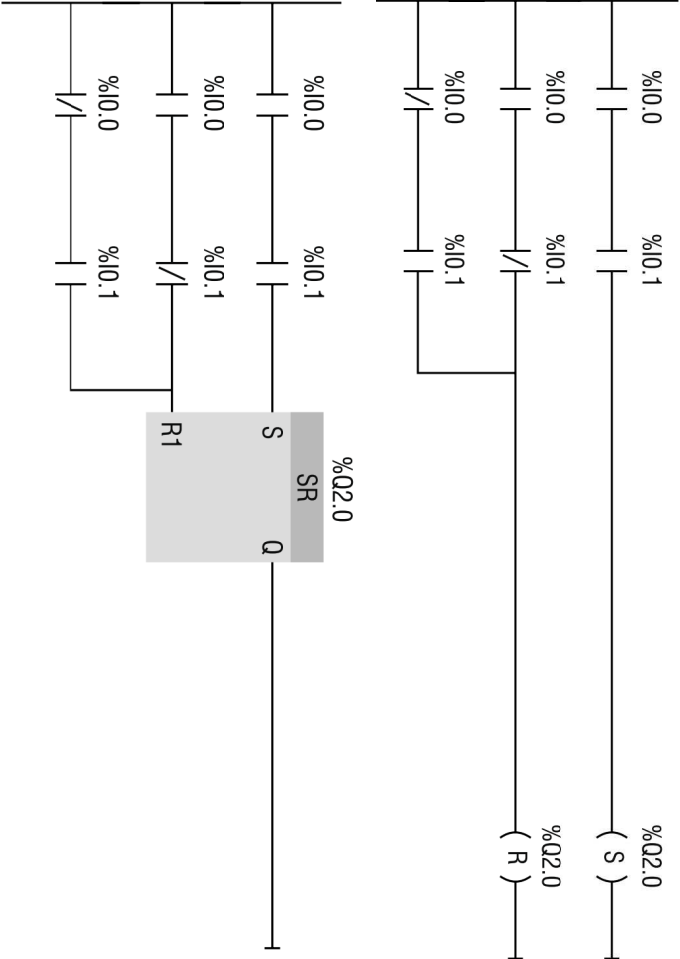
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ON/OFF –Set, Reset Coils



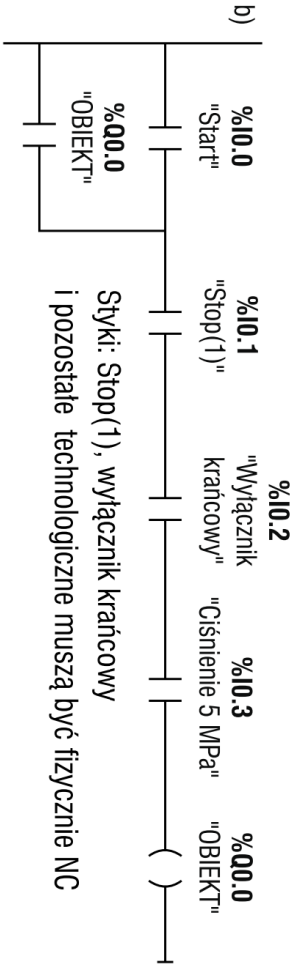
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Coils S, R and flip-flop SR



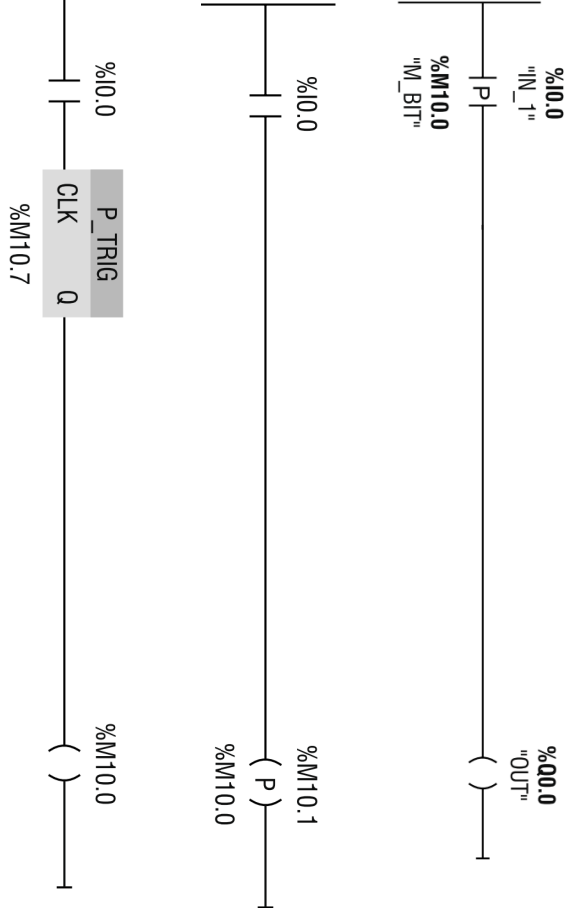
Wykorzystano ilustracje z książki:
J. Kwaśniewski „Sterowniki S7-1200
w praktyce inżynierskiej”

ON/OFF – diagram with self-
holding

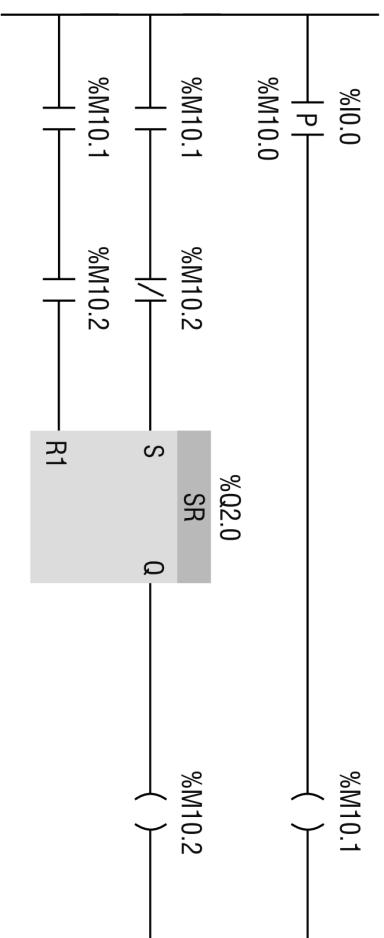


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Signal slope detection



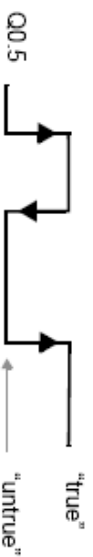
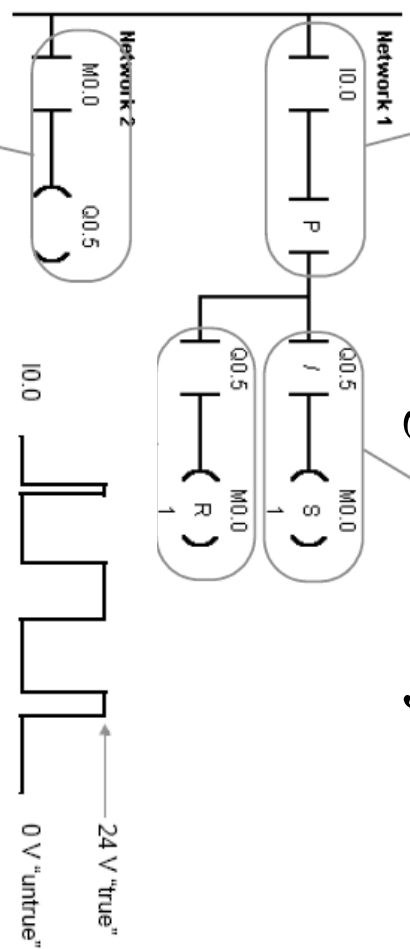
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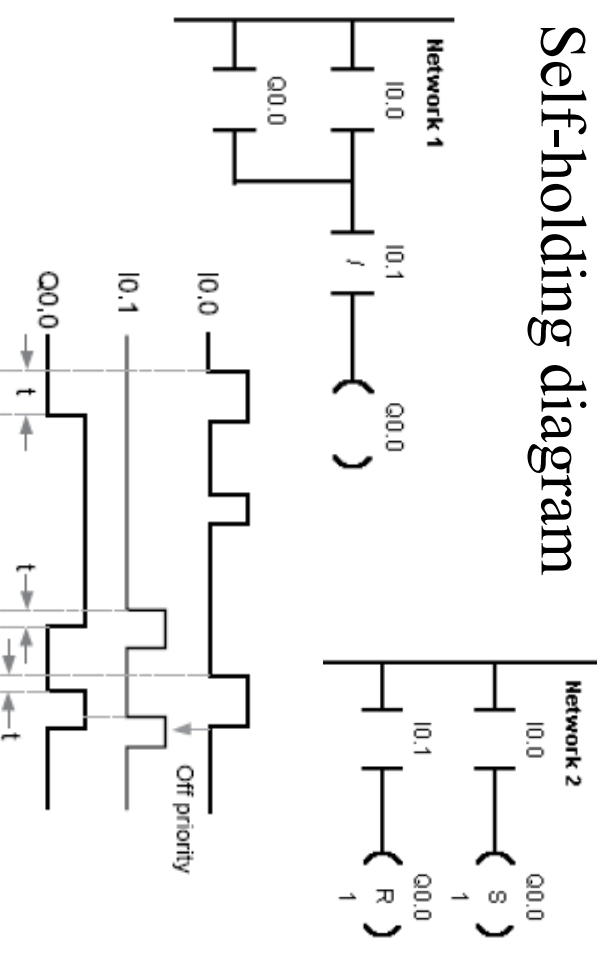
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Switching - auxiliary bit



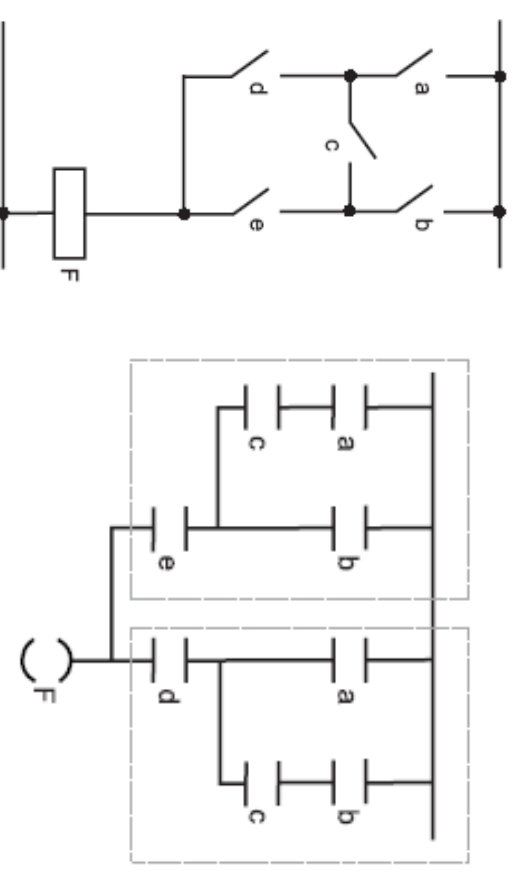
Self-holding diagram



t = time till the results of logic operations are transferred to the outputs (= response time).

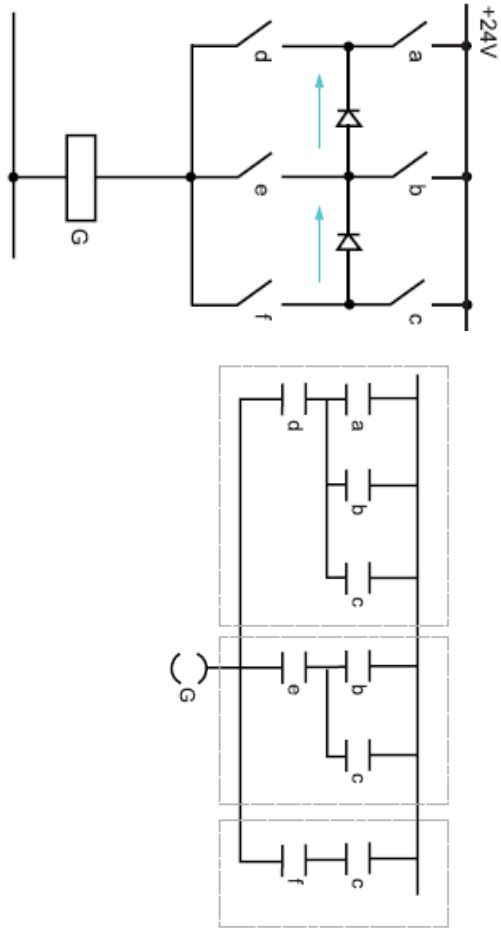
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Bridge connection



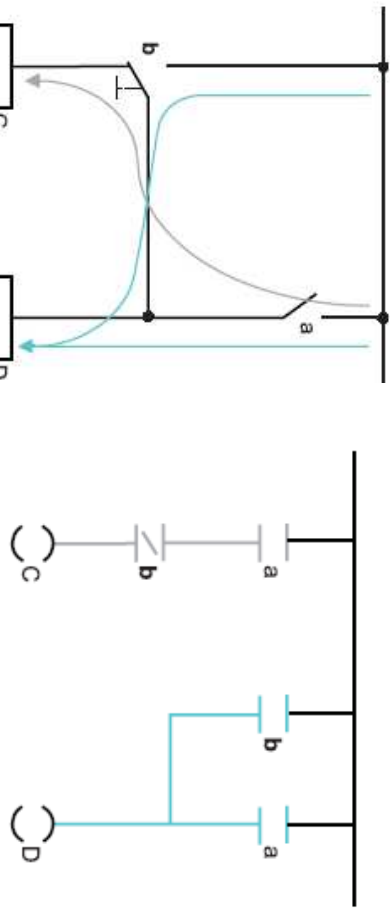
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Connection with diodes



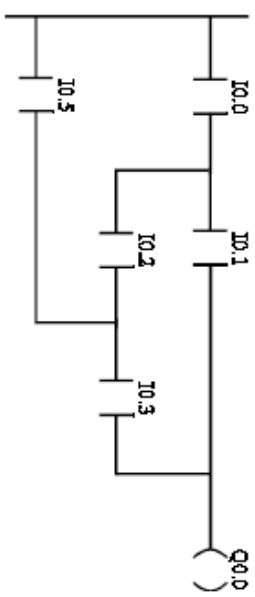
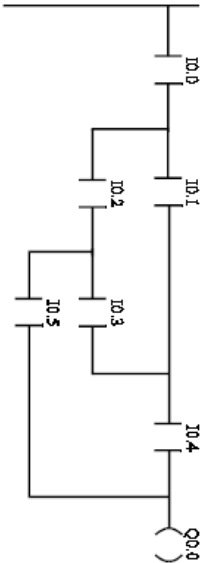
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Toggle switch



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Incorrect



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TON type timer

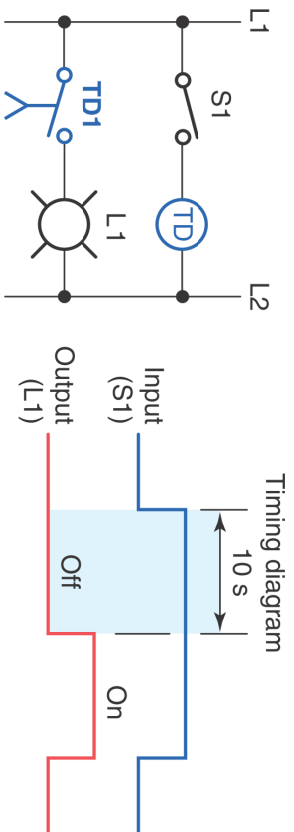


Figure 7-4 On-delay timer circuit that uses a normally open, timed closed (NOTC) contact.

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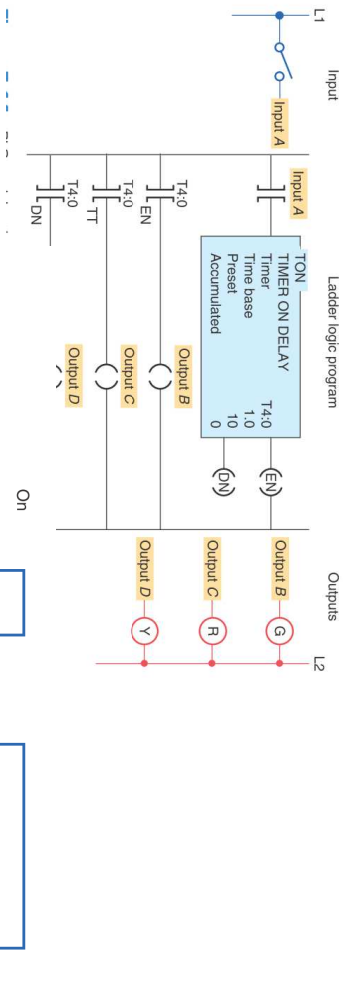
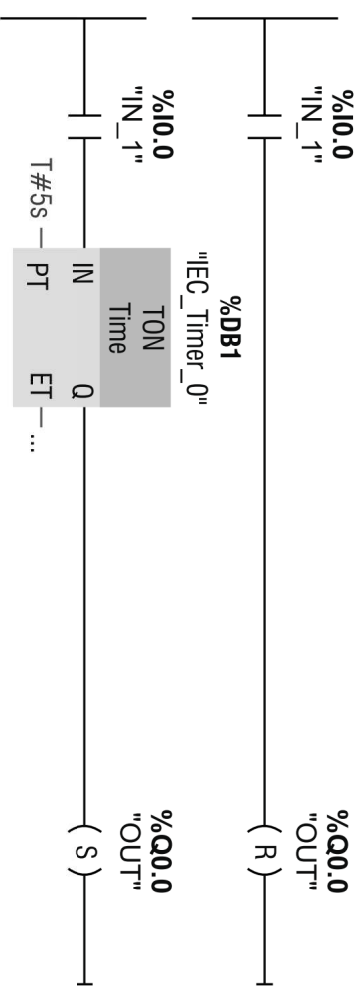


Figure 7-15 Timing diagram for an on-delay timer.

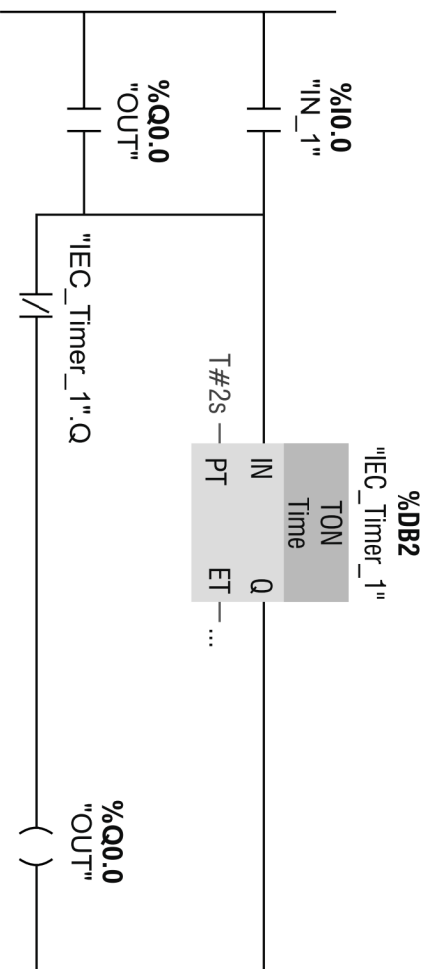
Delayed switching on



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Pulse with set time duration



TOF type timer

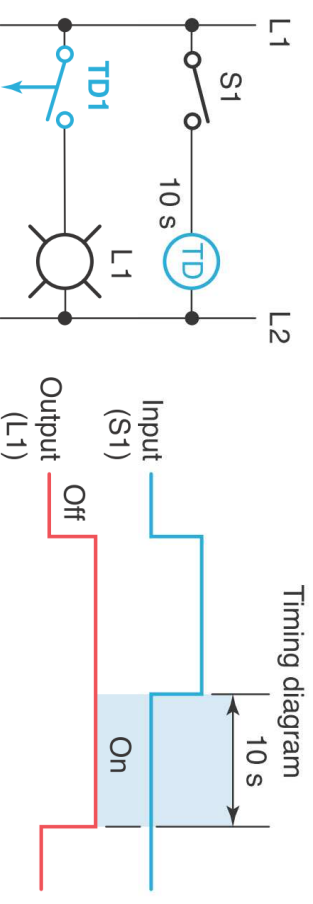
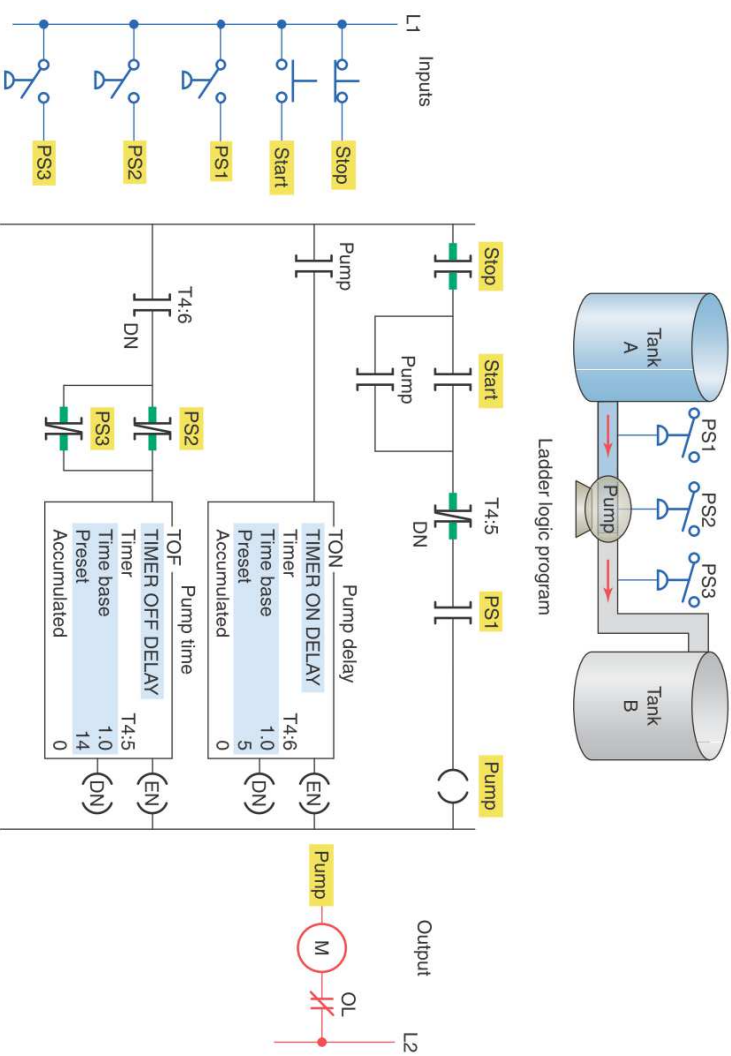


Figure 7-6 Off-delay timer circuit that uses a normally open, timed open (NOTO) contact.



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Switching on and off delay

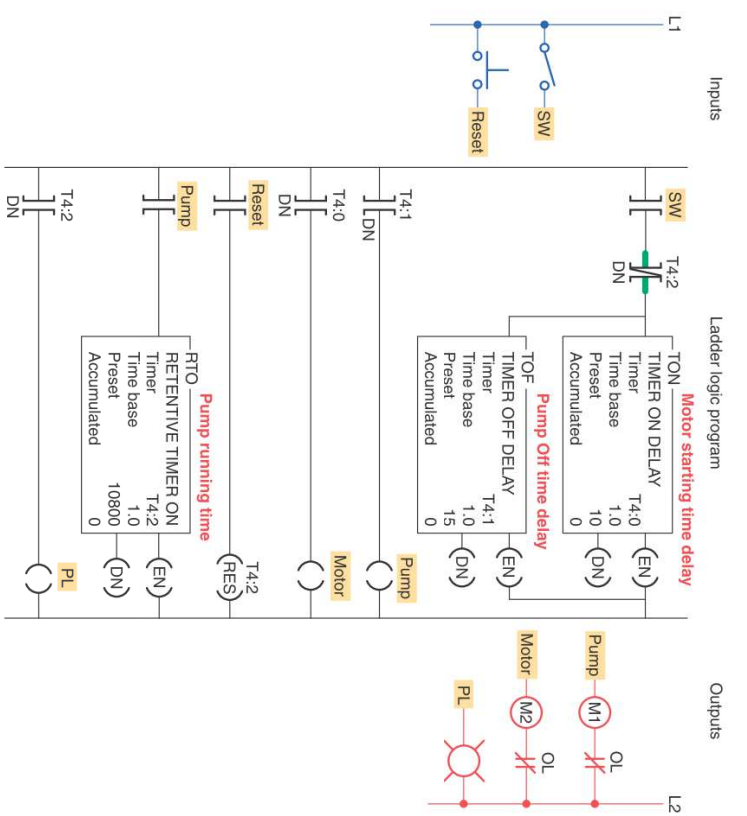
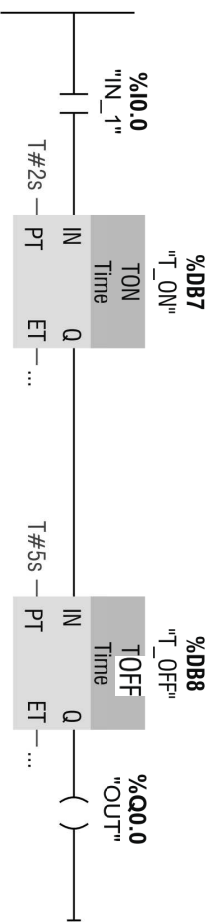


Figure 7-29 Bearing lubrication program.

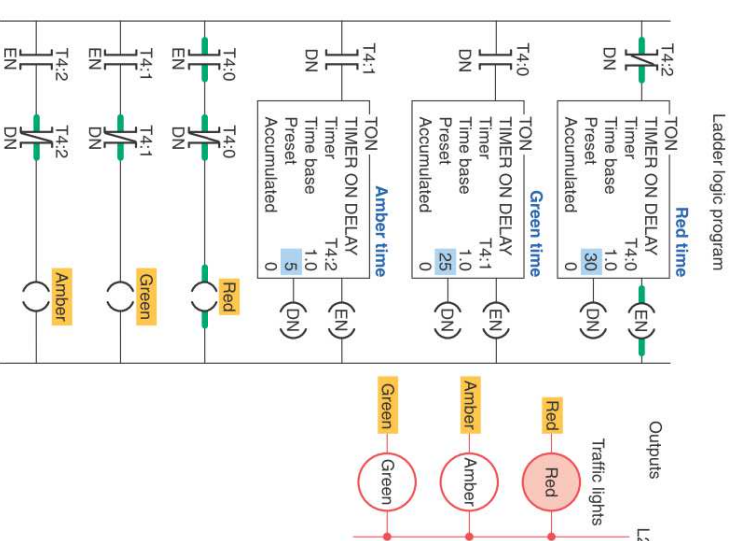
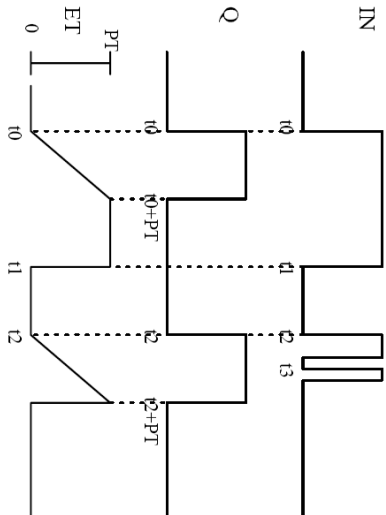


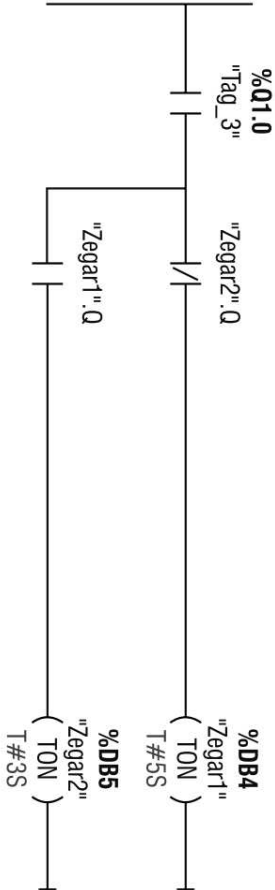
Figure 7-34 Control of traffic lights in one direction.

TP type timer

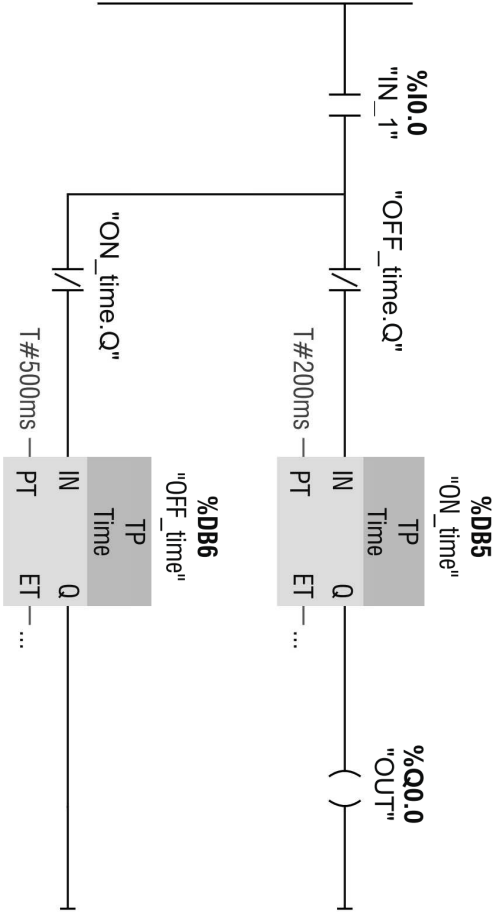


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Rectangular wave

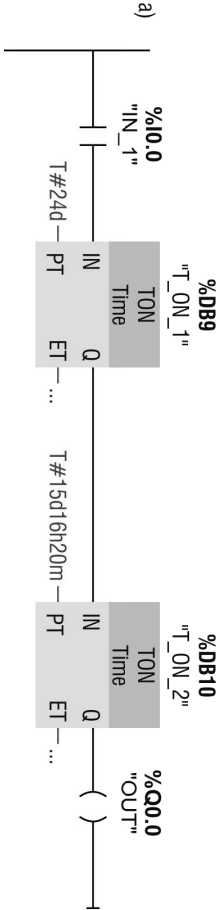


Rectangular wave



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Cascade connection of timers

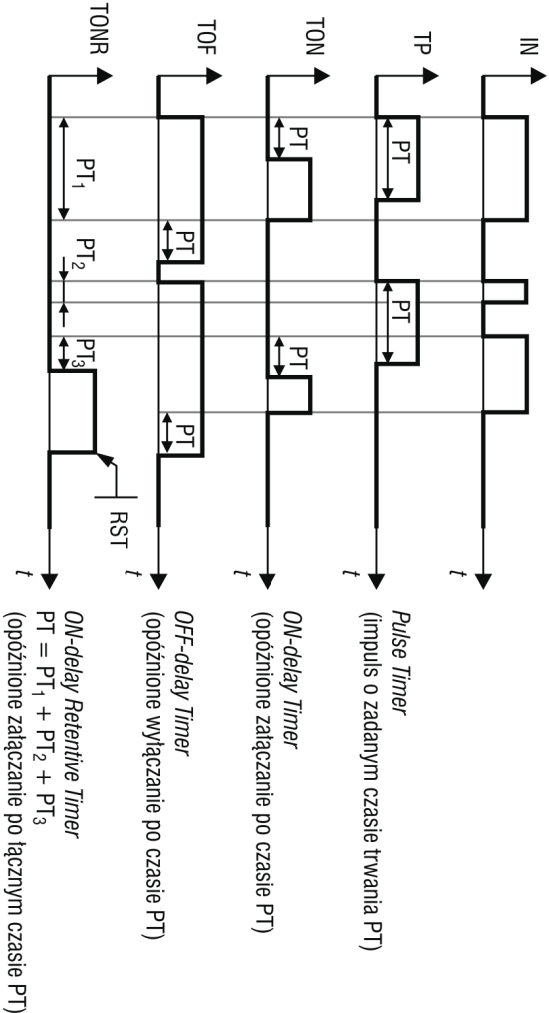


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Timer	Timing diagram
TP: Pulse timer The TP timer generates a pulse with a preset width time.	
TON: ON-delay timer The TON timer sets output Q to ON after a preset time delay.	

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Timer	Timing diagram
TOF: OFF-delay timer The TOF timer resets output Q to OFF after a preset time delay.	
TONR: ON-delay Retentive timer The TONR timer sets output Q to ON after a preset time delay. Elapsed time is accumulated over multiple timing periods until the R input is used to reset the elapsed time.	

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